VD detector protection

Extending the BDE PLC system

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Fermilab

BDE already has the PLC and panel design

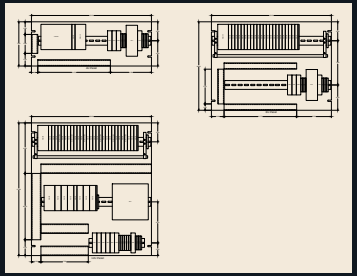


Diagram. TPC protection panel drawings.

Beckhoff PLC, distributed EtherCAT I/O and hardware supply inhibits.

VD allocation: 10 BDE auxiliary racks and 20 PL506 supplies for 80 WIECs.

Panel drawings, parts lists, power calculations and the Ignition front end already exist.

Proposal: extend this design to VD, using industrial parts we can maintain and expand.

BDE = bottom drift electronics; PLC = programmable logic controller; WIEC = warm interface electronics crate. Sources: TPC panel package, VD rack workbook and June 2026 SC presentations.

Put the VD PLC in the CUC, with I/O near the equipment

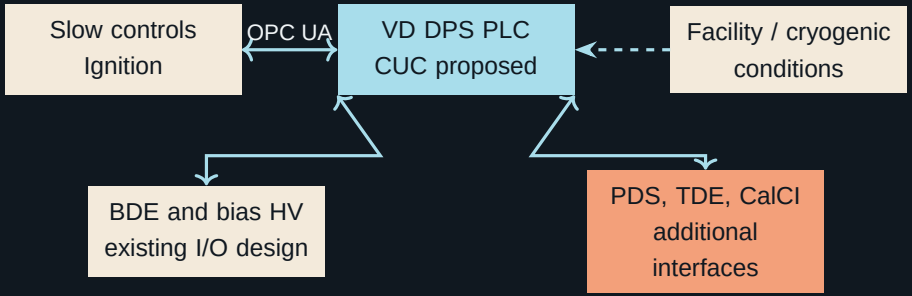


Diagram. Proposal: place the central PLC beside the other PLC racks in the CUC; retain distributed I/O.

CUC = Central Utility Cavern. Dashed link: facility signals need a selected hardware or software path. PDS = photon detection; TDE = top drift electronics; CalCI = calibration and cryogenic instrumentation. HD can use an extension or a separate PLC later.

Add the VD interfaces that the BDE system does not cover

Known VD scope	Proposed DPS addition
PDS: 80 PoF boxes; 58 LV feeds	Local PoF readout/control; supply inhibits and trip feedback
TDE: 20 racks	Crate-status exchange and external inhibit interface
CalCI	Level, temperature and optical-source status; operation permits
BMS / cryogenics	Selected facility conditions used by DPS; monitoring shared with SC

The 80 PoF boxes produce 1,280 monitoring values. Local readout handles that volume; the protection permits determine the DPS I/O.

Working VD inventory and ICD matrix; exact trip scope remains subsystem-specific. BMS = building management system. SDSTA and LBNF/DUNE have separate BMSs.

Protection must keep working when SC is unavailable

SC

Operate and monitor

Commands, sequencing, displays and history. An operating request cannot override a protection trip.

DPS / local protection

Protect equipment

PLC permits and device protection remain effective through an SC outage. Reset and re-enable are separate actions.

The PX3 setup already has local fallback. The VD prototype will exercise it together with PLC protection and SC recovery.

Proposed control model. The PX3 guide uses Ignition for normal protection and PDU fallback when Ignition is unreachable; this is a concrete integration test with Geoff and the rack team.

Test the full protection path before buying the additions

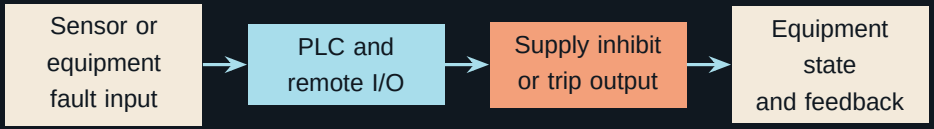


Diagram. Use the BDE vertical-slice work and G271 equipment for representative VD paths.

Measure detection through completed shutdown against the equipment's allowed time. Exercise SC, fieldbus and power loss, then reset and restart.

No critical sub-ms function has been identified so far. The simulation prepares load and fault cases; the bench measures hardware behaviour.

Take the VD hardware to review in June 2027

Work before review	Result for procurement
Adapt the BDE design	VD panels, added I/O, CUC fieldbus routes and power allocation
Test the interfaces	Trip-time, failure and recovery results; SC/PDU integration
Prepare purchases	Drawings, parts and cable lists, costs, lead times and spares

After readiness review: order the hardware and install it in the site sequence.

Possible facility-interface need: summer 2027. Full VD response-matrix review: early 2028. The early matrix work supplies requirements that affect hardware purchases.

I will lead the VD hardware work with a control engineer

Proposed contribution	Work
Manuel + control engineer	VD design, SURF interfaces and procurement package
Terri and Trevor	BDE panels, PLC/I/O implementation and bench work
Geoff + Manuel	SC commands, PDU protection and recovery
Vivian + consortia / DPS	Equipment requirements and ICD updates

Support needed: control-engineering time and access to the existing test stands.

SURF work: inspection maps are ready; access is being coordinated with Rachel, with Dale Curran identified as an SDSTA integration contact.

Backup: VD rack space already allocated to DPS

Rack type	Count	DPS equipment / connections
BDE auxiliary	10	12U per rack; eight WIEC connections per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U per rack for the DPS interface to ISEG supplies

The auxiliary racks contain the WIEC I/O; the bias racks contain the ISEG inhibit outputs.

May 2025 VD rack workbook; June 2026 BDE layout. U is rack height. These are design allocations; the bias-HV sheet is a first estimate.

One BDE I/O group serves eight WIECs

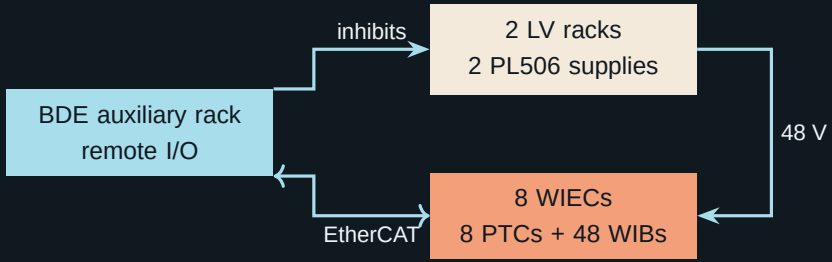


Diagram. Ten groups: 20 PL506 supplies, 80 WIECs and 480 WIBs.

The auxiliary rack also connects the fans, heaters and temperature sensors.

WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. June 2026 layout; fan/heater wiring omitted here.

Panel drawings and the Ignition front end are available

TPC protection panel designs

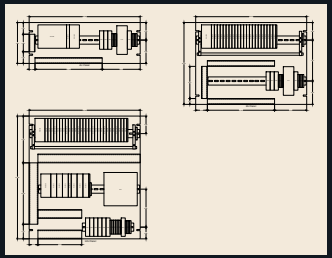
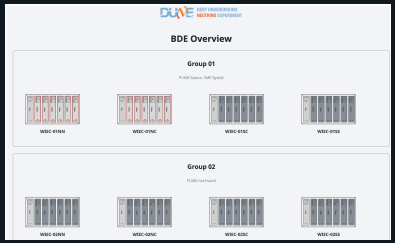


Diagram. Central PLC, bias I/O and LV/fan/heater panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

TPC reference: EDMS 2401090 v4, sheet 2; component names from the 2023 parts list.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. Quantities describe the first-pass panel layout.

VD power estimates for the BDE racks

Equipment	Per rack	VD basis
PL506 input	2.25 kW	20 racks: about 45.1 kW for the WIEC supplies
Auxiliary DPS input	About 1.11 kW	10 racks: about 11.1 kW; eight WIECs per rack
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The auxiliary estimate scales the recorded 831 W from six to eight WIECs: $831 \times 8/6 = 1{,}108$ W.

Planning estimates from the May 2025 workbook. Linear scaling follows its VD note; it is not a measured load. Supply capacity is separate from AC demand.

The PX3 rack-protection setup already defines shutdown

June 2026 setup guide	Documented behaviour
Ignition connected	Ignition reads sensors and switches outlets over SNMP
Ignition unreachable	PX3 local rules handle smoke, water and temperature; the network-switch outlet stays on
After a fallback trip	Power is restored manually or through Ignition

Test this setup together with the PLC's protection permits and trip feedback.

DUNE-doc-32655 v8, pp. 3–6 and 14–16. The PDU fallback is documented; the PLC permit integration is the proposed extension.

The VD inventory covers the other subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	42 DAPHNE boards and 8 calibration boxes; 58 LV feeds from MPOD modules
Power over fibre (PoF)	80 laser boxes; 640 temperature and 640 voltage readings; compact PLC interface proposed
TDE	20 mezzanine racks; power supplies and readout-crate management
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

VD hardware lists and SC architecture. Proposed routing: monitoring to Ignition through native drivers and OPC UA; protection permits and fault status exchanged with the local equipment.

The BMS drawing identifies 20 control-panel symbols

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Inspection planned with Rachel; Dale Curran is the SDSTA integration contact identified for this work.

U-FD-M-901 rev12, 14 February 2025; marked 4850/4910 maps and survey register. Symbols are drawing references; field observations are pending.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 allocation: rack positions, distinct from PLC panels and remote-I/O stations. Includes the listed reserved and spare positions.